

Mairin O'Shaughnessy

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EDUCATION

Massachusetts Institute of Technology

Candidate for B.S. in Computer Science and Molecular Biology

GPA: 4.9/5.0

Relevant Coursework: Intro to Algorithms, Software Construction, Computation Structures, ML for Molecular Engineering, Genetics, Applied Cryptography (Graduate)

Cambridge, MA

Expected May 2027

EXPERIENCE

Software Intern, Applied Invention

Jun 2026 – Aug 2026

- Built a computer vision pipeline for automated pipette tip counting to support lab automation workflows
- Evaluated locally-hosted LLM inference on Oracle Cloud Infrastructure GPUs against cloud-based alternatives, and designed an experimental framework to benchmark model performance, cost, and context limits for data science tasks
- Restructured system prompts by switching from listed tool schemas to tool search, improving inference performance and reducing cost

Software Engineering Intern, TriMin Systems

Jun 2025 – Aug 2025

- Resolved form-validation and input-handling bugs in the JavaScript frontend, replacing user-facing error messages with internal handling for expected edge cases
- Patched a broken access control vulnerability involving URL manipulation; expanded unit test coverage from 27% to 40%

Software Engineering Intern, TriMin Systems

Jun 2024 – Aug 2024

- Increased unit test coverage from near-zero to 27% using JUnit and Mockito, embedding test-driven development into the team workflow and surfacing bugs earlier in the development cycle

UNDERGRADUATE RESEARCH

Semantic Segmentation for Graphite Phase Classification — Bazant Group, MIT

Sep 2024 – May 2025

- Implemented a PyTorch neural network for colorimetric segmentation of graphite particles, resolving class imbalance through weighted categorical loss and LAB color space representation

Computer Vision for Plant Density Analysis — Nepf Lab, MIT

Jan 2024 – May 2024

- Developed OpenCV-based models to predict stem count from images of marsh plants
- Designed an efficient image-cropping method for test data, reducing processing time

LEADERSHIP

Tutoring Committee Co-Director, MIT Eta Kappa Nu

Jul 2026 – Present

- Organizing the undergraduate tutoring program for MIT's largest academic department

Mathematics Tutor

Oct 2025 – Present

- Tutoring high school and college students in Advanced Geometry, Introduction to Probability, and Math for Computer Science

UNDERGRADUATE PROJECTS

Hush — Creating Video Games (6.4570)

Fall 2025

- Collaborated with a team of six to design and build a medieval-themed side-scrolling video game using an Agile framework, built around the theme of "Playful Audio"
- Designed and integrated animation assets using Aseprite, Unity Engine, and C#

StackBot — Robotic Manipulation (6.4210)

Fall 2025

- Implemented the perception module of a full-stack system for manipulating a virtual 6-DoF robotic arm to stack blocks of varying sizes
- Used the Drake robotics library to crop and filter 3D environment point clouds and estimate object positions via geometric pose estimation; developed additional geometric techniques to estimate object sizes

AWARDS AND HONORS

- FIRST Dean's List Finalist Award
- IEEE-HKN Honor Society — invited as part of the top 25% of MIT juniors in the EECS department